

Simple Linear Regression & Correlation Coefficient

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What Will We Learn Today?

No scary math - just simple steps!

By the end, you can:

- ✓ Draw a “best fit” line through data points
- ✓ Calculate the line equation by hand (with small numbers!)
- ✓ Understand what slope and intercept mean in IT
- ✓ Measure how strong a relationship is (correlation)
- ✓ Predict values for new data

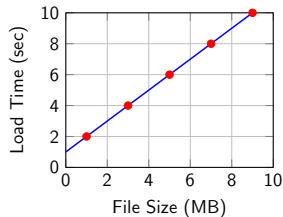


Figure: Simple example: bigger files take longer to load

What is Bivariate Analysis?

Studying relationships between TWO variables

Definition:

- **Bivariate** = "two variables"
- Analysis of how one variable changes when another changes
- Foundation for prediction and pattern discovery

Key Question:

"When X changes, what happens to Y ?"

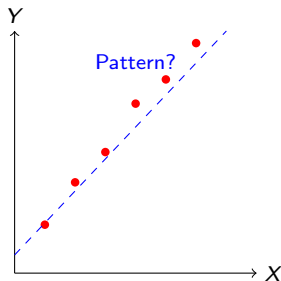


Figure: Scatter plot: visualizing bivariate data

IT Examples of Bivariate Pairs

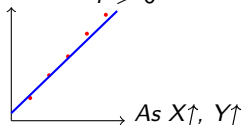
- **Cloud Computing**: CPU usage (X) $\leftrightarrow$ Response time (Y)
- **Software Testing**: Code complexity (X) $\leftrightarrow$ Bug count (Y)
- **Network Admin**: Packet size (X) $\leftrightarrow$ Transfer time (Y)

Types of Bivariate Relationships

What patterns can we observe?

Positive Linear

$$r > 0$$

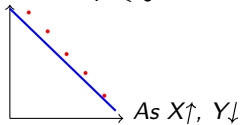


As $X \uparrow$, $Y \uparrow$

Example: More users $\rightarrow$ Higher load

Negative Linear

$$r < 0$$

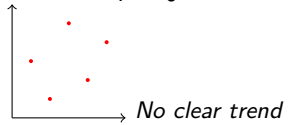


As $X \uparrow$, $Y \downarrow$

Example: More cache $\rightarrow$ Lower latency

No Linear Pattern

$$r \approx 0$$



No clear trend

Example: Team size $\rightarrow$ Code quality?

Important Distinction

- **Linear:** Points roughly follow a straight line
- **Non-linear:** Points follow a curve (quadratic, exponential, etc.)
- **No relationship:** Points scattered randomly

Simple linear regression only models LINEAR relationships!

Visualizing Bivariate Data: Scatter Plots

Your first step in any bivariate analysis

Why plot first?

- See the **overall pattern** (linear? curved? clusters?)
- Spot **outliers** that could distort calculations
- Check for **non-constant spread** (heteroscedasticity)
- Build intuition before running formulas

IT Example: API Performance

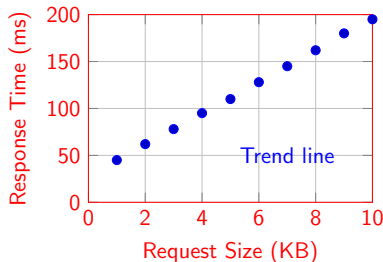


Figure: Clear positive linear pattern

What to look for:

- 1 **Direction**: Upward or downward trend?
- 2 **Form**: Straight line or curve?
- 3 **Strength**: Tight cluster or wide scatter?
- 4 **Unusual points**: Any outliers?

Rule of Thumb

Never skip the plot!

Calculating regression on unexamined data is like driving without looking.

What is Linear Regression?

Formal definition made simple

Core Idea

Finding the straight line that best describes
how one variable predicts another

Mathematical Model:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Components:

- Y : **Dependent variable** (what we predict)
- X : **Independent variable** (what we know)
- β_0 : **Intercept** (Y when $X=0$)
- β_1 : **Slope** (change in Y per unit X)
- ε : **Error** (unexplained variation)

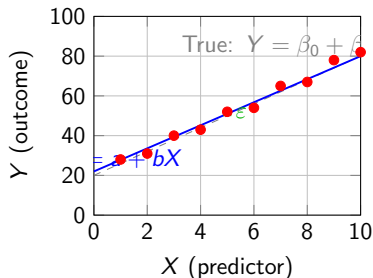


Figure: Data + True line + Fitted line + Error

When to Use Simple Linear Regression

Appropriate use cases in IT

✓ Good Fit When:

- You have **two continuous variables**
- Scatter plot shows **roughly linear pattern**
- Goal is **prediction** or **quantifying relationship**
- You want a **simple, interpretable model**

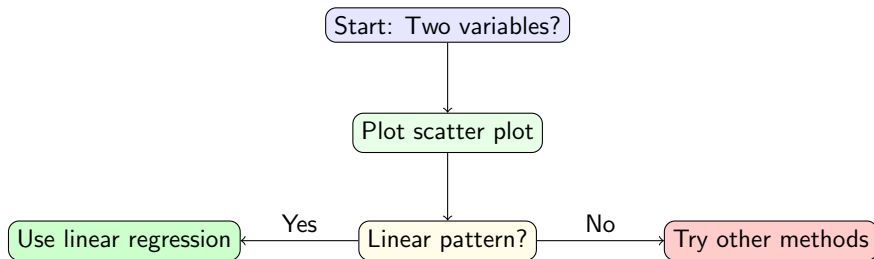
IT Scenarios:

- Predict deployment time from code size
- Model energy consumption vs. CPU load
- Estimate support tickets from user growth
- Forecast storage needs from data growth rate

✗ Avoid When:

- Relationship is clearly **curved or complex**
- Y is **categorical** (use logistic regression)
- You have **multiple predictors** (use multiple regression)
- Data has **strong time dependence** (use time series)
- Goal is **proving causation** (use experiments)

Decision Flowchart



The "Best Fit" Line: Least Squares Principle

How do we choose the best line?

Intuition: Find the line that is **closest to all data points**. What does "closest" mean?

- For each point, measure **vertical distance** to the line
- This distance is called the **residual**: $e_i = Y_i - \hat{Y}_i$
- Square each residual (to handle positive/negative)
- **Minimize the sum of squared residuals**

$$\min_{a,b} \sum_{i=1}^n \underbrace{(Y_i - (a + bX_i))^2}_{\text{squared residual}}$$

Why Square the Residuals?

- Prevents positive/negative errors from canceling out
- Penalizes large errors more heavily (desirable)
- Leads to nice mathematical properties (unique solution)

The Basic Formula

Just one simple equation

$$Y = a + bX$$

What each part means:

- Y = What we want to predict (the answer)
Example: App load time in seconds
- X = What we know (the input)
Example: Image file size in MB
- a = Starting point (when $X = 0$)
Example: Base load time even for tiny files
- b = How much Y changes when X increases by 1
Example: Each extra MB adds how many seconds?

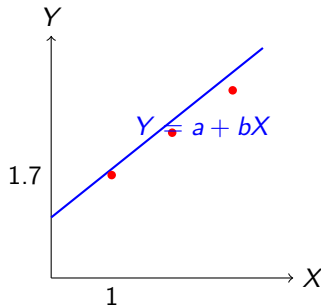


Figure: Visual: Line through data points

Manual Example 1: App Loading Time

Tiny dataset - easy to calculate by hand!

Scenario: You test an app with different image sizes:

Test	Size X (MB)	Time Y (sec)	$X \times Y$	X^2
1	1	2	2	1
2	2	3	6	4
3	3	5	15	9
4	4	6	24	16
SUM	10	16	47	30

Step 1: Calculate averages

$$\bar{X} = 10 \div 4 = 2.5$$

$$\bar{Y} = 16 \div 4 = 4$$

Step 2: Calculate slope b

$$b = \frac{\sum XY - n\bar{X}\bar{Y}}{\sum X^2 - n\bar{X}^2} = \frac{47 - 4(2.5)(4)}{30 - 4(2.5)^2} = \frac{47 - 40}{30 - 25} = \frac{7}{5} = 1.4$$

Step 3: Calculate intercept a

$$a = \bar{Y} - b\bar{X} = 4 - 1.4(2.5) = 4 - 3.5 = 0.5$$

Final Equation: $Y = 0.5 + 1.4X$

Interpreting Our Result

What does $Y = 0.5 + 1.4X$ mean?

In plain English:

- **Intercept $a = 0.5$:**
Even a 0 MB file takes 0.5 seconds to load (app startup time)
- **Slope $b = 1.4$:**
Each additional MB adds 1.4 seconds to loading time
- **Prediction:**
For a 5 MB image:
 $Y = 0.5 + 1.4(5) = 0.5 + 7 = 7.5$ seconds

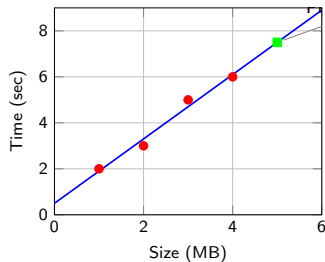


Figure: Our line + prediction for 5 MB

Try It Yourself!

If an image is 2.5 MB, what's the predicted load time?

Answer: $Y = 0.5 + 1.4(2.5) = 4$ seconds ✓

Correlation Coefficient (r) - The Simple Version

How strong is the relationship?

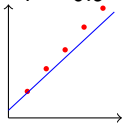
$$r = \frac{\text{How X and Y move together}}{\text{How much X and Y vary separately}}$$

What r tells us:

- $r = 1$: Perfect positive relationship (as $X \uparrow$, $Y \uparrow$ exactly)
- $r = -1$: Perfect negative relationship (as $X \uparrow$, $Y \downarrow$ exactly)
- $r = 0$: No linear relationship
- r between -1 and 1: Strength of linear pattern

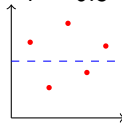
Strong +

$r \approx 0.9$



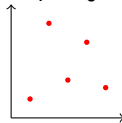
Weak

$r \approx 0.3$



None

$r \approx 0$



Manual Example 2: Calculating r by Hand

Using our app loading data

Recall our data (now with Y^2 column):

Test	X (MB)	Y (sec)	$X \times Y$	X^2	Y^2
1	1	2	2	1	4
2	2	3	6	4	9
3	3	5	15	9	25
4	4	6	24	16	36
Sum	10	16	47	30	74

Formula (simplified version):

$$r = \frac{n \sum XY - (\sum X)(\sum Y)}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$$
$$r = \frac{4(47) - (10)(16)}{\sqrt{(4(30) - (10)^2)(4(74) - (16)^2)}} = \frac{188 - 160}{\sqrt{(120 - 100)(296 - 256)}} = \frac{28}{\sqrt{(20)(40)}} \approx 0.99$$

Interpretation

$r = 0.99$ means: **Very strong positive relationship!**

File size explains almost all the variation in loading time.

When NOT to Use Simple Regression

Important limitations for IT applications

Watch out for:

- **Curved patterns:** If data looks like a curve, not a line
- **Outliers:** One weird point can break your line
- **Time data:** Server logs often have patterns over time
- **Multiple causes:** Real IT problems usually have many factors

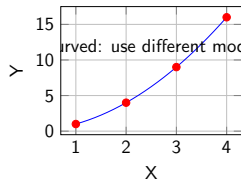


Figure: Example: Quadratic pattern

Golden Rule

Always plot your data first!

If points don't roughly follow a straight line, simple regression may mislead you.

Correlation $\neq$ Causation: IT Examples

Critical thinking for future professionals

- **Example 1:** More GitHub commits $\leftrightarrow$ more bugs found
Reality: Both increase during active development phases
- **Example 2:** Higher server CPU $\leftrightarrow$ more user complaints
Reality: High CPU is a symptom, not the cause of slowness
- **Example 3:** More IT staff $\leftrightarrow$ fewer system outages
Reality: Could be causal, or both caused by bigger budget

How to think about it

- 1 Does X happen *before* Y? (timing)
- 2 Is there a logical mechanism? (theory)
- 3 Could a third factor cause both? (confounding)
- 4 Does it make sense in the IT context? (domain knowledge)

Safe approach

Use regression for **prediction** (“if X, then likely Y”)

Use experiments for **causation** (“changing X causes Y to change”)

Quick Reference: Formulas Cheat Sheet

For manual calculations

Given data pairs (X_i, Y_i) for $i = 1$ to n :

Quantity	Formula
Mean of X	$\bar{X} = \frac{\sum X}{n}$
Mean of Y	$\bar{Y} = \frac{\sum Y}{n}$
Slope	$b = \frac{\sum XY - n\bar{X}\bar{Y}}{\sum X^2 - n\bar{X}^2}$
Intercept	$a = \bar{Y} - b\bar{X}$
Correlation	$r = \frac{\sum XY - n\bar{X}\bar{Y}}{\sqrt{(\sum X^2 - n\bar{X}^2)(\sum Y^2 - n\bar{Y}^2)}}$
Prediction	$\hat{Y} = a + bX_{\text{new}}$

Pro Tips

- Make a table with columns: X , Y , XY , X^2 , Y^2
- Calculate sums first, then plug into formulas
- Use a calculator for square roots, but keep fractions exact when possible
- Always check: Does your slope sign make sense? (positive/negative)

Summary: Key Takeaways

What to remember

Simple Linear Regression:

- Equation: $Y = a + bX$
- b = change in Y per 1-unit change in X
- a = Y value when $X = 0$
- Use for prediction within data range

Correlation Coefficient:

- r measures linear relationship strength
- Range: -1 to +1
- r^2 = percent of Y variation explained by X
- **Does NOT prove causation!**

IT Student Checklist

- 1 ✓ Plot data before analyzing
- 2 ✓ Use small, clean datasets for manual practice
- 3 ✓ Interpret slope/intercept in real-world terms
- 4 ✓ Question causality claims
- 5 ✓ Know when to use software for big data

Practice Problem 1: API Response Time

Try it yourself - small numbers!

Scenario: Testing an API with different request sizes:

Request	Size X (KB)	Time Y (ms)
A	1	50
B	2	70
C	3	90
D	4	110

Your tasks:

- 1 Calculate $\bar{X}$ and $\bar{Y}$
- 2 Find slope b and intercept a
- 3 Write the prediction equation
- 4 Predict time for a 5 KB request
- 5 Estimate correlation r (is it strong?)

Hint: Use the table method

Add columns for $X \times Y$, X^2 , Y^2 , then use the formulas!

Practice Problem 2: Cloud Storage Cost

Another simple manual example

Scenario: Monthly cloud cost vs. data stored:

Month	Data X (GB)	Cost Y (\$)
Jan	10	15
Feb	20	25
Mar	30	35
Apr	40	45

Questions:

- 1 Without calculating: What do you expect for b (slope)?
- 2 Calculate the regression equation
- 3 Predict cost for 25 GB
- 4 What is r ? What does it tell you?
- 5 Is this relationship causal? Why/why not?

Real-world thinking

Even with perfect correlation, ask: Does storing more data *cause* higher cost, or is there a pricing rule? (Answer: Pricing rule - so yes, causal here!)

Try These at Home!

Extra practice with IT scenarios

Problem A: Website Traffic

Visitors (X) vs. Server Load (Y):

Hour	1	2	3	4
Visitors (100s)	2	4	6	8
Load (%)	30	45	60	75

Find equation, predict load for 500 visitors.

Problem B: Code Review Time

Lines of Code (X) vs. Review Minutes (Y):

PR	A	B	C	D
Lines	10	20	30	40
Minutes	15	25	35	45

Calculate r . Is review time strongly related to code size?

Problem C: Critical Thinking

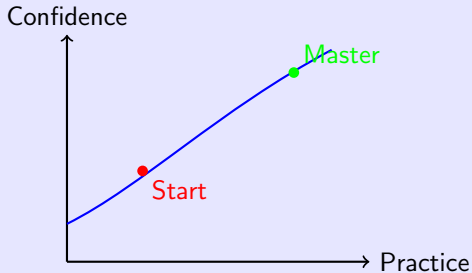
If $r = 0.95$ between “number of alerts” and “system downtime”, does reducing alerts cause less downtime?

Explain.

You Got This!

Simple Regression for IT

Start with small data, understand the story, then scale up



Questions? Ask anytime!

"The best way to learn statistics is to do statistics."